

Here is the breakdown of the **optimized formulations** from your dataset, along with the **master batch recipe** and **step-by-step laboratory preparation procedure** grounded in the research sources.

## 1. Optimized Formulation Comparison

From the formulation optimization dataset (bio\_spray\_optimized\_formulations.csv), three distinct formulations were modeled and evaluated based on droplet size, viscosity, active load, and predicted follicular penetration 1:

Parameter, Formulation A (Top Penetration), Formulation B (Highest Score), Formulation C (Ultra-Lightweight)

Carrier & Actives, Squalane + Pea Sprout Nanovesicles 1, Jojoba Oil + Resveratrol Nanovesicles 1, Fractionated Coconut (MCT) + Coffee Extract 1

Droplet Size, 45.0 nm 1, 85.0 nm 1, 150.0 nm 1

Viscosity, 18.5 cP 1, 24.0 cP 1, 15.0 cP 1

Active Load, 5.0% 1, 7.5% 1, 3.0% 1

Follicular Penetration, 80.08% 1, 67.54% 1, 53.39% 1

Optimization Score, 55.04 1, 56.27 1, 35.70 1

- **Formulation A** achieves the highest follicular penetration (80.08%) and ultra-small droplet size (45 nm), making it the ideal benchmark for the **micro-mist bio-spray** 1.

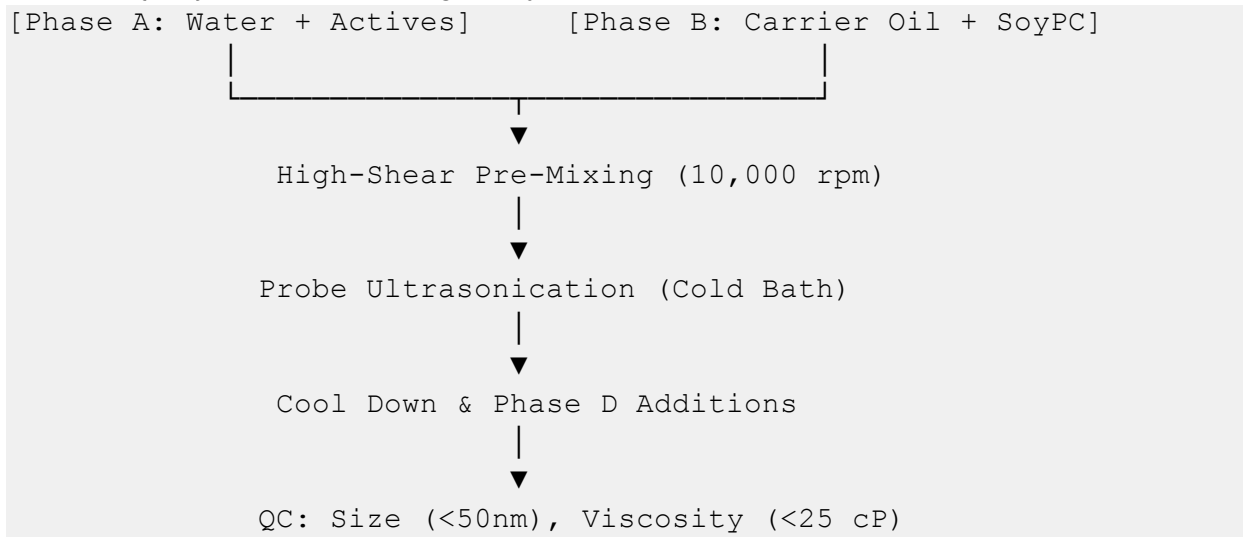
## 2. Master Formulation Phase Breakdown (100 g Batch)

To produce an Oil-in-Water (O/W) nanoemulsion spray with low viscosity ( $<25 \text{ cP}$ ) 1, 2, the phase components are organized as follows:

- **Phase A: Aqueous Phase (80.0–85.0%)**
- Deionized Water (q.s. to 100%) 3, 4
- Botanical Actives: Pea Sprout Extract or water-dispersible botanical bio-actives (1.0–3.0%) 1, 5
- Chelating Agent / Hydrator: Sodium Phytate / Glycerin (0.5–1.0%) 3, 6
- **Phase B: Oil & Lipophilic Active Phase (5.0–10.0%)**
- Low-viscosity Carrier Oil: Squalane, Jojoba Oil, or Crodamol/MCTs (3.0–5.0%) 1, 7
- Encapsulated Bio-Actives: Resveratrol Nanovesicles or Coffee seed diterpenes/tocopherols (1.0–2.5%) 1, 8, 9
- **Phase C: Emulsifier & Surfactant System (3.0–5.0%)**
- Non-Ionic Emulsifier Blend: Soy Phosphatidylcholine (SoyPC) / Lecithin combined with Tween 80 or Labrasol (SOR = 1:2 to 1:3 ratio) 10-13
- **Phase D: Cool-Down & Preservation (1.0–2.0%)**
- Broad-Spectrum Green Preservative: Benzyl Alcohol / Ethylhexylglycerin (0.8–1.0%) 4, 14
- Antioxidant: Tocopherol (Vitamin E, 0.2–0.5%) 8, 15

### 3. Step-by-Step Preparation Procedure

The manufacturing process utilizes a **high-energy emulsification method** (high-shear mixing followed by ultrasonication or high-pressure homogenization) to reduce droplet sizes to  $<50\text{ nm}$  without requiring heavy waxes or occlusive oils 12, 14, 16:



#### Step 1: Phase Preparation

1. **Oil Phase (Phase B + Phase C):** Combine carrier oil (Squalane/Jojoba) 1 with the lipid emulsifier blend (SoyPC/Tween 80) 11, 12 in a clean glass beaker. Gently mix at room temperature or low heat ( $40\text{--}45^\circ\text{C}$ ) until completely uniform.
2. **Aqueous Phase (Phase A):** In a separate vessel, dissolve the water-soluble botanical extracts (Pea Sprout extract) 1 and chelating agents in deionized water under moderate stirring 17.

#### Step 2: Coarse Pre-Emulsification

1. Slowly pour the Oil Phase into the Aqueous Phase under constant mechanical agitation 14.
2. Process the combined mixture using a **high-shear rotor-stator homogenizer** (e.g., Ultra-Turrax) at **6,000–10,000 rpm for 3 to 5 minutes** 12, 18, 19. This generates a coarse emulsion with micro-scale oil droplets 20.

#### Step 3: Nanoemulsification (Droplet Reduction)

1. Subject the coarse emulsion to **ultrasonic probe homogenization** (e.g., 70% amplitude, 5:5 pulse cycle for 10–15 minutes) 12, 21.
2. **Temperature Control:** Maintain the vessel in an **ice/cold-water bath** during ultrasonication to prevent thermal degradation of heat-sensitive botanical antioxidants (such as resveratrol and flavonoids) 12, 22.
3. **Alternative Industrial Method:** Pass the pre-emulsion through a **high-pressure homogenizer** at 500–1,000 bar for 3–5 cycles to achieve droplet sizes below 50 nm 12, 23, 24.

#### Step 4: Cool-Down & Final Compounding

1. Allow the nanoemulsion to cool to below  $(30^{\circ}\text{C})$  20.
2. Add **Phase D** (preservative system and tocopherol) under gentle magnetic stirring (500 rpm) for 10 minutes until homogeneous 14, 25.

#### 4. Quality Control & Characterization Standards

Before bottling into micro-mist spray actuators, the batch must meet the following benchmark specifications:

- **Droplet Size & Polydispersity Index (PDI):** Confirm hydrodynamic droplet diameter  $(20\text{--}80\text{ nm})$  and monodisperse distribution  $(\text{PDI} < 0.25)$  using Dynamic Light Scattering (DLS) 26, 27.
- **Viscosity:** Verify low viscosity  $(<25\text{ cP})$  at  $(25^{\circ}\text{C})$  on a rotational rheometer to ensure complete atomization in micro-mist sprayers without clogging nozzle orifices 2, 28.
- **Scalp Compatibility pH:** Test and adjust acidity to **pH 5.0–5.5** (matching physiological scalp microenvironment) 26.
- **Physical Stability Testing:** Centrifuge sample at 4,000 rpm for 30 minutes and perform 6 freeze-thaw cycles  $(-120^{\circ}\text{C})$  to  $(+40^{\circ}\text{C})$  to verify resistance against creaming, phase separation, or Ostwald ripening 26, 29, 30.

 *Would you like me to draft a printable PDF technical spec sheet with these exact formulation ratios and safety parameters for your lab team or CMO partner?*